Yuen Chen

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EDUCATION

University of Illinois at Urbana-Champaign — Expected May 2028
Ph.D. Candidate in Computer Science, advised by Prof. Hari Sundaram and Prof. Han Zhao
Research Topics: Causal Machine Learning, Trustworthy Machine Learning

University of California, Berkeley — Graduated December 2022
B.A. Applied Mathematics & Statistics — GPA: 3.93/4.00

University of Copenhagen — February 2022 - June 2022
Exchange Student in Computer Science and Mathematical Science — GPA: 10.5/12

Irvine Valley College — Graduated May 2020
Associate in Science in Mathematics — GPA: 3.94/4.0

RESEARCH EXPERIENCE

Applied Research Intern | AI Foundation, Capital One — June 2026 - Aug 2026
Supervisors: Yulun Wu & Nam Nguyen
- Investigated in-context learning and tabular foundation model pretraining for causal discovery
- Manuscript in preparation; planned submission to ICML 2027

Applied Research Intern | AI Foundation, Capital One — June 2025 - August 2025
Mentor and Manager: Rizal Fathony & Nam Nguyen
- Developed BRIDGE, an end-to-end Transformer–GNN pipeline combining user event histories with relation graphs
- Designed controlled experiments across five datasets to identify when joint training improves predictions
- Improved friend recommendation quality by training user activity and social network models together

Research Intern | Empirical Inference, Max Planck Institute for Intelligent Systems — February 2023 - July 2023
Supervisors: Prof. Bernhard Schölkopf & Zhijing Jin
- Developed the "CLadder" benchmark to assess formal causal reasoning ability in Large Language Models (LLMs)
- Proposed a prompting strategy to elicit multi-step formal causal reasoning in LLMs
- Conducted experiments analyzing the impact of semantic representations on LLM performance

Undergraduate Researcher | Language, Reasoning and Education Lab, ETH Zürich — August 2023 - December 2022
Supervisors: Prof. Mrinmaya Sachan & Zhijing Jin
- Conducted research on experimental design to optimize experimental tables with orthogonal arrays

Undergraduate Researcher | University of Copenhagen — February 2022 - June 2022
Supervisor: Prof. Yevgeny Seldin
- Conducted research to find an optimal algorithm for multi-armed bandits problems with time-based switching costs
- Reviewed literature in online learning, gaining insights into current methodologies and trends

PEER-REVIEWED PUBLICATIONS

Why Freeze the Encoder? A Controlled Study of End-to-End Sequence–Graph Learning for User Behavior. *(LoG 2026)*
Yuen Chen, Yulun Wu, Samuel Sharpe, Igor Melnyk, Nam H. Nguyen, Furong Huang, C. Bayan Bruss, Rizal Fathony

CauSciBench: Can LLMs Automate Causal Inference in Real-World Scientific Research? *(ICML 2026)*
Sawal Acharya, Terry Jingchen Zhang, Andrew Kim, Rahul Babu Shrestha, Xianlin Sun, Pepijn Cobben, Maximilian Mordig, Jacob T. Emmerson, Anahita Haghighat, Furkan Danisman, **Yuen Chen**, Clijo Jose, Andrei Ioan Muresanu, Justin Cui, Jiarui Liu, Yahang Qi, Punya Syon Pandey, Yinya Huang, Bernhard Schölkopf, Zhijing Jin.

CausalDetox: Causal Head Selection and Intervention for Language Model Detoxification. *(Findings of ACL 2026)*
Yian Wang, **Yuen Chen**, Agam Goyal, Hari Sundaram

Moment Alignment: Unifying Gradient and Hessian Matching for Domain Generalization. *(UAI 2025)*
Yuen Chen, Haozhi Si, Guojun Zhang, Han Zhao

Breaking Bad Tokens: Detoxification of LLMs Using Sparse Autoencoders. *(EMNLP 2025)*
Agam Goyal, Vedant Rathi, William Yeh, Yian Wang, **Yuen Chen**, Hari Sundaram

Causally Testing Gender Bias in LLMs: A Case Study on Occupational Bias. *(Findings of NAACL 2025)*
Yuen Chen*, Vethavikashini Chithrra Raghuram*, Justus Mattern*, Rada Mihalcea, Zhijing Jin.

Analyzing the Role of Semantic Representations in the Era of Large Language Models. *(NAACL 2024)*
Zhijing Jin*, **Yuen Chen***, Fernando Gonzalez Adauto*, Jiarui Liu, Jiayi Zhang, Julian Michael, Bernhard Schölkopf, Mona Diab.

CausalCite: A Causal Formulation of Paper Citations. *(Findings of ACL 2024)*
Ishan Kumar Agrawal*, Zhijing Jin*, Ehsan Mokhtarian, Siyuan Guo, **Yuen Chen**, Mrinmaya Sachan, Bernhard Schölkopf.

CLadder: Assessing Causal Reasoning in Language Models. *(NeurIPS 2023)*
Zhijing Jin*, **Yuen Chen***, Felix Leeb*, Luigi Gresele*, Ojasv Kamal, Zhiheng Lyu, Kevin Blin, Fernando Gonzalez, Max Kleiman-Weiner, Mrinmaya Sachan, Bernhard Schölkopf.

PREPRINTS

CausaLab: A Scalable Environment for Interactive Causal Discovery Toward AI Scientists. *(arXiv 2026)*
Junlin Yang, Dylan Zhang, Xiangchen Song, Qirun Dai, Xiao Liu, **Yuen Chen**, Aniket Vashishtha, Jing Shi, Chenhao Tan, Hao Peng

DIVEBATCH: Accelerating Model Training Through Gradient-Diversity Aware Batch Size Adaptation. *(arXiv 2025)*
Yuen Chen, Yian Wang, Hari Sundaram

HIGHLIGHTED PROJECTS

Time Series Analysis on Semiconductor Processing Tools — Fall 2022
- Developed a machine learning model to predict the performance of wafer production process
- Condensed 560k+ data into 216 data by Fast Fourier Transform, sinusoidal regression, and quadratic regression
- Achieved 87% prediction accuracy on wafer metrology with a linear regression model

Offline Evaluation of Bandit Algorithms — Spring 2022
- Evaluated bandit algorithms with importance-weighted losses on "R6B Yahoo! Front Page Today Module User Click Log Dataset"
- Investigated the performance of UCB1, EXP3, and random strategy compared to the theoretical performance bound
- Achieved 2.6x better performance on EXP3 algorithm than the theoretical performance lower bound

Medical Images Segmentation — Spring 2022
- Implemented U-Net with PyTorch to segment the blood vessels on photographs of the retina
- Trained the neural network with sample-splitting and output segmented images for the test image data
- Achieved <10% binary cross-entropy loss after 40 training epochs

Representation Learning and Generative Modelling on MNIST Dataset — Spring 2022
- Performed dimensionality reduction on MINIST dataset using PCA, Autoencoder (AE), and Variational Autoencoder (VAE)
- Optimized binary cross-entropy loss of AE from 0.005 to 0.002 and evidence lower bound loss of VAE from 500+ to 200
- Synthesized new image data by sampling from the latent space of VAE

R Packages Text Analysis — Spring 2021
- Performed frequency text analysis by regular expression and visualized the outcome by R data visualization tool
- Built a web application with R-shiny that demos the analysis of a distribution of R packages' title length

PROFESSIONAL & LEADERSHIP EXPERIENCE

Statistics Course Reader | University of California, Berkeley — September 2022 - December 2022
- Assisted and graded 800+ students on homework, labs, exams weekly in Stat 20: Introduction to Statistics
- Co-managed 8 sections of lectures with Prof. Andrew Bray and other course staff on course content

Math Tutor | Irvine Valley College — August 2019 - May 2022
- Conducted 1-on-6 weekly sections and hosted office hours for 200+ students weekly
- Designed weekly problem sets for teaching uses in fields of trigonometry, calculus, and differential equations
- Earned high satisfaction in teaching from students in Fall 2021 and Spring 2022

Commissioner | Student Government of Irvine Valley College — August 2018 - September 2019
- Coordinated with 6 other committees on planning and advertising campus events
- Allocated $800,000 in funds to student organizations, intercurricular programs, and scholarships

*Equal Contribution